

Harkamal Singh

Data Scientist

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Junior Data Scientist with 1.5+ years of hands-on project experience in data science and machine learning. Focused on turning data into clear business decisions by choosing right approach for each problem and building solutions that are practical, measurable and explainable.

Education

Conestoga College | Diploma, Computer Programming *Waterloo, Ontario | 09.2024 - 04.2026*

Relevant Coursework: Data Structure, Object-Oriented Programming, Database, Software Development, Artificial Intelligence & Machine Learning, Statistics & Probability

Skills

Programming & Querying: Python, SQL

Data Science & Statistics: pandas, NumPy, EDA, Feature Engineering, Statistics & Probability, A/B Testing, Experimentation, Causal Inference, Time Series Forecasting

Machine Learning: scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch

Database & Data Tools: MySQL, MongoDB, Redis, DuckDB

ML Engineering & MLOps: FastAPI, Docker, MLflow, ZenML, Git, GitHub Actions, pytest

Projects

Movie Recommendation A/B Testing & Uplift Modeling | *Python, SciPy, scikit-learn, pandas*

- Designed a randomized experiment across **28,000 users** comparing personalized and popularity-based recommendations.
- Measured a **+0.85 pp CTR lift (p=0.017)** and used segmentation and uplift modeling to identify returning users as the strongest rollout segment.

LocalMart – Retail Promotion Intelligence | *Python, Machine Learning, Recommendation Systems, Streamlit*

- Built an end-to-end retail recommendation system to generate relevant product and promotion recommendations for local customers.
- Developed a **5-km location-aware recommendation feature** to surface relevant Walmart products and promotions near the user, improving the practical relevance of recommendations.

Flight Pulse – AI Flight Delay Analyst | *Python, SQL, APIs, Streamlit, Gemini*

- Built an analytics application combining flight, weather, and external data to investigate delays and operational conditions.
- Developed an interactive dashboard with AI-assisted explanations for flight status and delay patterns.

Fighting Churn – Customer Churn Prediction | *Python, pandas, scikit-learn, Machine Learning*

- Built a classification workflow to identify customers at risk of churn using customer and behavioral features.
- Performed data preprocessing, feature engineering, model evaluation, and analysis to support customer-retention decisions.

Toronto Climate & Time-Series Analysis | *Python, pandas, Statistics, Time Series*

- Analyzed Toronto climate data from **1953–2023** to measure long-term and seasonal temperature trends.
- Estimated an annual warming trend of approximately **0.22°C per decade** and compared trends across all four seasons.

End-to-End ML Pipeline | *Python, ZenML, MLflow, Docker, FastAPI*

- Developed a modular machine-learning pipeline covering data ingestion, preprocessing, model training, and evaluation.
- Applied ML engineering practices using **ZenML and MLflow** for reproducible pipelines and experiment tracking.